

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.2: Ecology — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.2: Ecology
Driving Question	DRIVING QUESTION: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Where Have the Fish Gone? Launching Our Investigation into Ecosystems	A Kenyan river that once had many fish now yields few, while algae and aquatic weeds grow thickly and the water sometimes smells unpleasant. The river is still full of life.	An ecosystem is a community of living organisms interacting with each other and with non-living factors in an area. Ecological terms such as habitat, niche, population, community, biome, biomass, carrying capacity and ecological succession help describe how the river works.	Fish being old or lacking food cannot alone explain the booming algae, weeds and smell. The river is an ecosystem where organisms and the environment, including human activity, interact, so changes in one part can affect others. We still need to learn how organisms are adapted to their habitats and what causes these changes.
Lesson 2 Who Lives Where? Habitat Classification and Adaptations in a Kenyan River and Beyond	We read about xerophytes, mesophytes, hydrophytes, halophytes, desert animals and aquatic animals and noted the conditions of each habitat and the adaptations of Kenyan organisms such as Euphorbia, maize, water hyacinth, mangroves, camels and tilapia.	Organisms are classified according to the habitat in which they live, and each group has structural and physiological adaptations (e.g. waxy cuticle and spines, air spaces, salt glands, gills, water-conserving kidneys) that help it survive water, salt, oxygen and temperature conditions.	In the river, conditions such as extra nutrients and warm water favour algae and aquatic weeds (hydrophytes), while fish with gills need enough dissolved oxygen, so changes in the environment can help some organisms and harm others. We still need to find out how many organisms live there and what the characteristics of populations are.
Lesson 3 Counting Life: Characteristics of Populations and Direct Counting in the School Compound	We marked two 10 m x 10 m areas in the school compound, counted every individual of one organism (such as grass tufts or weeds), and recorded the number, area and density for each site. The sites differed in population size and density.	A population is all the individuals of one species in an area. Population density is the number of individuals per unit area (number / area). Direct counting is suitable when the population is small and individuals are easy to identify and count.	Sites differed because of factors such as sunlight, soil, water and trampling. Direct counting works for the school compound but not for fish hidden in a river, so other methods are needed. We now ask: Is the distribution of grass uniform throughout the school field?

Lesson 4 Quadrat Sampling: Is the Distribution of Grass and Weeds Uniform Across the School Field?	We used 1 m x 1 m quadrats placed randomly in different zones of the school field and counted grass and weed plants. We saw that the numbers per quadrat were very different in sunny, shaded, damp and trampled areas.	Plants are not spread evenly in a habitat. Population density is the number of individuals per unit area, and quadrat sampling lets us estimate population size without counting every plant. Light, water, soil and trampling affect the size of plant populations.	Just as weeds were dense in some parts of the school field because of local conditions, algae and aquatic weeds boom in parts of the river where conditions, such as nutrients from human activity, favour them. This helps explain why the river is full of life but fish are disappearing, and leads us to ask how plant populations change as environmental conditions change.
Lesson 5 Belt Transect Across a Changing Habitat: How Does Plant Population Change as Conditions Change?	We stretched two parallel lines across the school compound (open field, shaded area, pathway, garden), placed quadrats at regular intervals and recorded plant abundance, light, soil moisture and trampling. Plant numbers differed from zone to zone (record the actual mean per square metre for each zone).	Population size changes across a habitat because conditions change. Abiotic factors (light, water, soil nutrients) and biotic factors (competition, grazing, human activity such as trampling and gardening) affect how many individuals can live in a place. A belt transect with quadrats is a sampling method used to estimate abundance.	Like the plants in the compound, populations in the river respond to changing conditions. Extra nutrients from farms and sewage can favour algae and weeds, while changes in oxygen, light and human activity can reduce fish. This helps explain why the river is still full of life but fish are disappearing. We still need to find out which factors matter most and how to estimate fish populations.
Lesson 6 How Many Fish Are Left? Capture-Recapture Simulation to Estimate Population Size	We marked 20 of 100 counters, mixed them, took a second sample of 20 and counted how many were marked (e.g. 5). Estimates varied between trials but were near the true value of 100.	Population size can be estimated by capture-recapture using: estimated population = (number initially marked x number in second sample) / number recaptured. More trials give more reliable estimates, and the method relies on assumptions such as thorough mixing and no migration, births or deaths.	We can estimate the number of fish in the river without counting every fish, and compare estimates over time to show whether the population is falling. The method shows how many fish there are but not why they are decreasing, so we must investigate the factors that affect population size next.
Lesson 7 Factors Affecting Population Size: Why Are the Fish Disappearing?	We observed that in the PhET simulation adding predators made the prey population fall and then rise and fall again, and reducing food made the population decline. We also read about biotic and abiotic limiting factors and saw in the river video that fish decreased while algae, weeds and a bad smell increased.	We learned that population size is controlled by biotic factors (predation, competition, disease, food) and abiotic factors (temperature, oxygen, light, nutrients, water level). Populations stop growing at the carrying capacity, and factors interact with each other.	Fish are probably disappearing because several factors work together: overfishing, predators and competition, plus extra nutrients that cause algae and weed growth, which reduces light and dissolved oxygen. The simulation cannot explain where the nutrients go, so we need to investigate how nitrogen and carbon affect living organisms.
Lesson 8 Nitrogen and Carbon Cycles: How Nutrients Move and Why Algae Boom in Our River	We observed that the nitrogen and carbon cycles move nutrients between air, soil, water and living things through fixation, nitrification, photosynthesis, respiration and decomposition, and that runoff of fertiliser, manure and sewage adds extra nutrients to the river, with more algae, weeds and bad smell.	We learned that decomposers and bacteria recycle nutrients, that too many nutrients cause algal bloom (eutrophication), and that when algae and weeds die, decomposers use up dissolved oxygen, which harms fish.	The river is losing fish because nutrient pollution from farms and settlements causes algae and weeds to boom; their decay by bacteria uses up oxygen and produces foul smell, so fish suffocate. New question: Does pollution affect organisms?

Lesson 9 Pollution, Garbage Rivers and Restoring Our River: The Final Explanation	We observed types of pollution at the shopping centre (plastics and polythene, smoke, exhaust fumes, oily and soapy runoff, blocked drains, noise) and found research evidence about garbage rivers and effluent in the Nairobi and Athi Rivers. Add your own observations and photos.	We learned that pollution adds nutrients and harmful substances to rivers, causing algal bloom, low dissolved oxygen, fish deaths and smell, and that burning waste and fossil fuels adds greenhouse gases that contribute to global warming and climate change. Add what you learned.	Final Explanation: Fish are disappearing because effluent, garbage and fertiliser runoff enrich the river with nitrates and phosphates, so algae and weeds boom; when they die, bacteria use up the oxygen and produce foul-smelling gases, so fish suffocate or leave. The community can restore the river by managing waste, treating effluent, planting riverbank vegetation and reducing pollution.
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ATTRIBUTION AND LICENSING

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